

Big Pharma: What really moves prices?

LLY, MRK, AMGN, PFE (2023–2025)

Data sources: yfinance, SEC/XBRL, sentiment analysis

[Collab link](#)

Goals of the project

In Big Pharma, stock prices are often assumed to react strongly to news sentiment (FDA decisions, clinical trials, M&A announcements).

Many analytics and trading approaches rely on NLP-based sentiment signals as direct price drivers.

However, Big Pharma is a highly regulated and expectation-driven sector, where:

- Key events are often priced in advance.
- Companies move together as a sector.
- Volatility and fundamentals may dominate short-term sentiment.

- Evaluate the real impact of news sentiment on stock prices.
- Determine which signals provide stronger and more stable insights.
- Benchmark sentiment against volatility, co-movement, and fundamentals.

Sector Context & Hypotheses



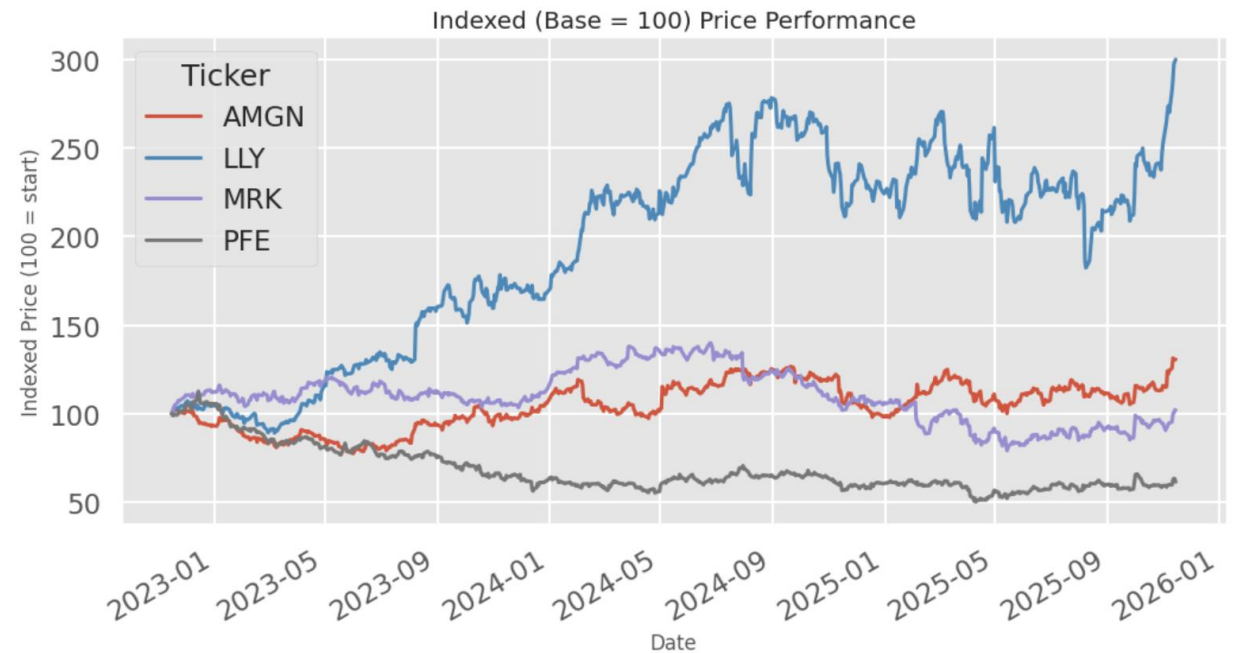
News sentiment $\neq$ price driver



Volatility, co-movement, fundamentals = stronger signals

Indexed Price Performance*

- Elly Lilly(LLY) outperforms: GLP-1 rerating
- Pfizer Inc.(PFE) underperforms: post-COVID normalization
- Merck & Co. (MRK) stable/defensive
- Amgen(AMGN) mid-pack



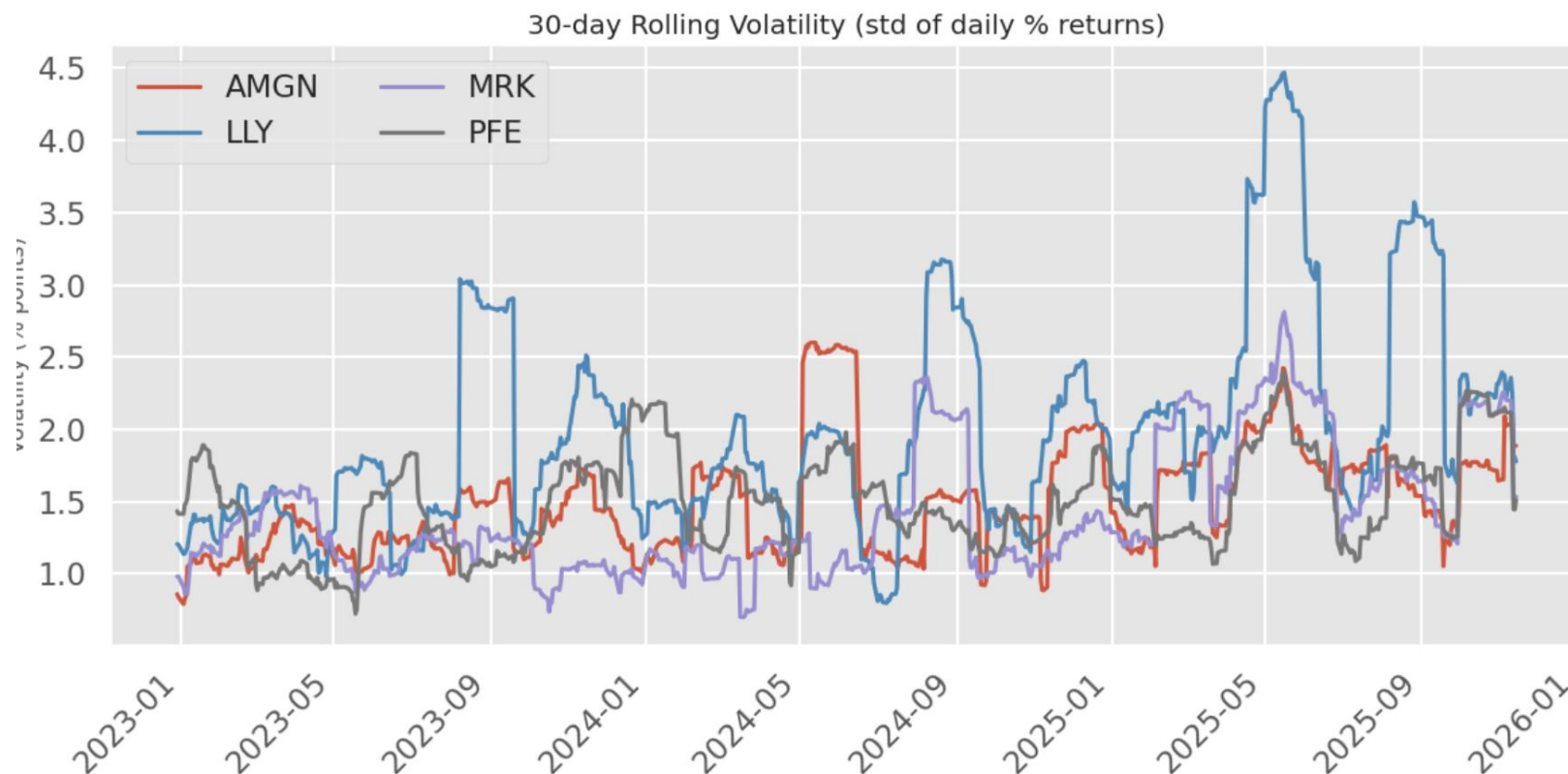
All prices are indexed to 100 at the start date to show relative performance rather than absolute price levels

*Comparison of relative stock price performance over time, indexed to a common starting point.

Volatility Trends

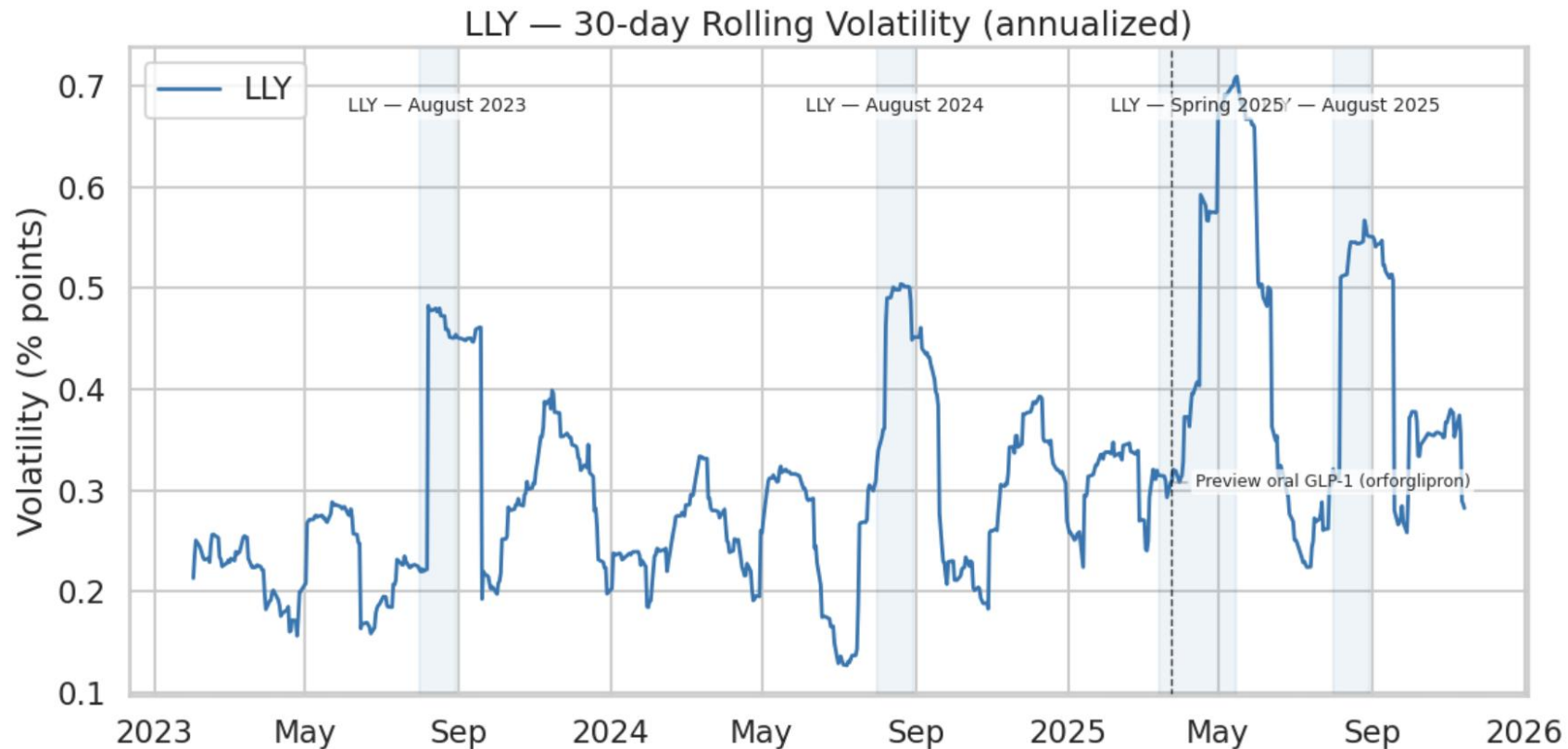
30-Day Rolling Volatility (% points)

- Volatility spikes cluster around earnings, clinical, FDA, legal events
- LLY shows event-driven spikes
- MRK/AMGN stable
- PFE spikes but no trend signal from sentiment



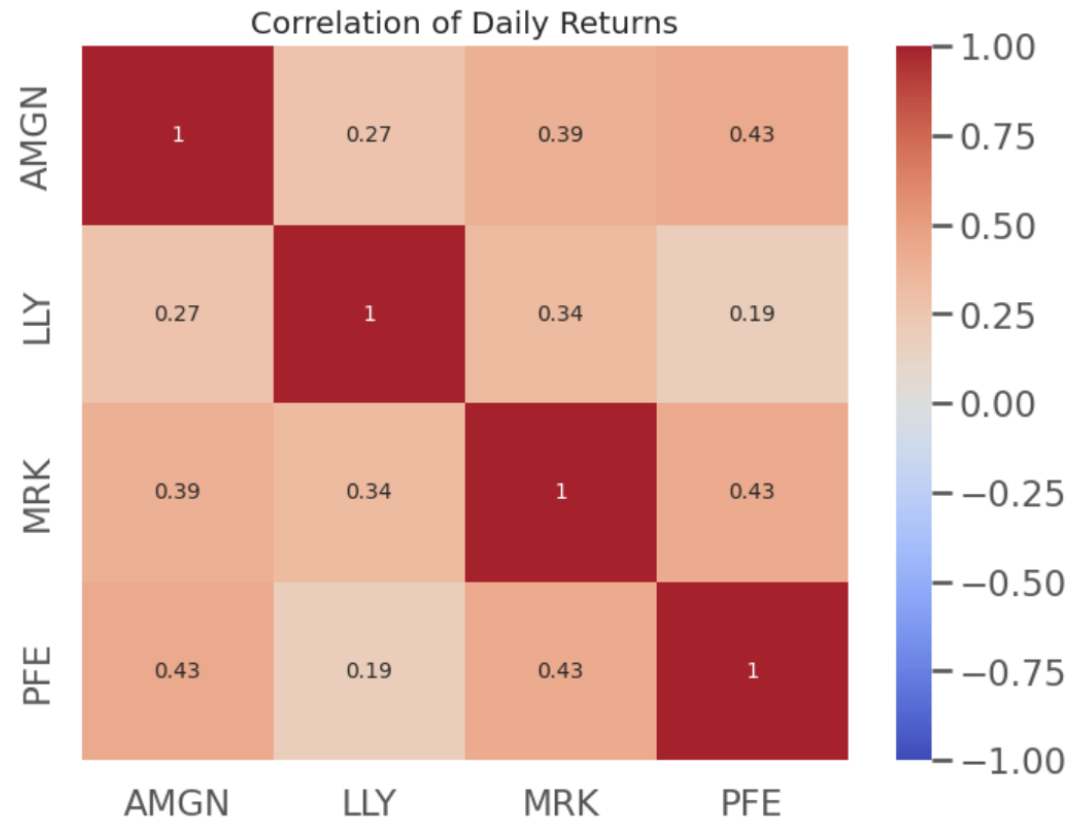
LLY Event Windows

- Aug 2023/2024/2025: Q2 earnings
- Spring 2025: Orforglipron Phase 3
- Summer 2024: Kisunla approval + Medicare coverage
- Summer 2025: Kisunla label update + ATTAIN-1 results



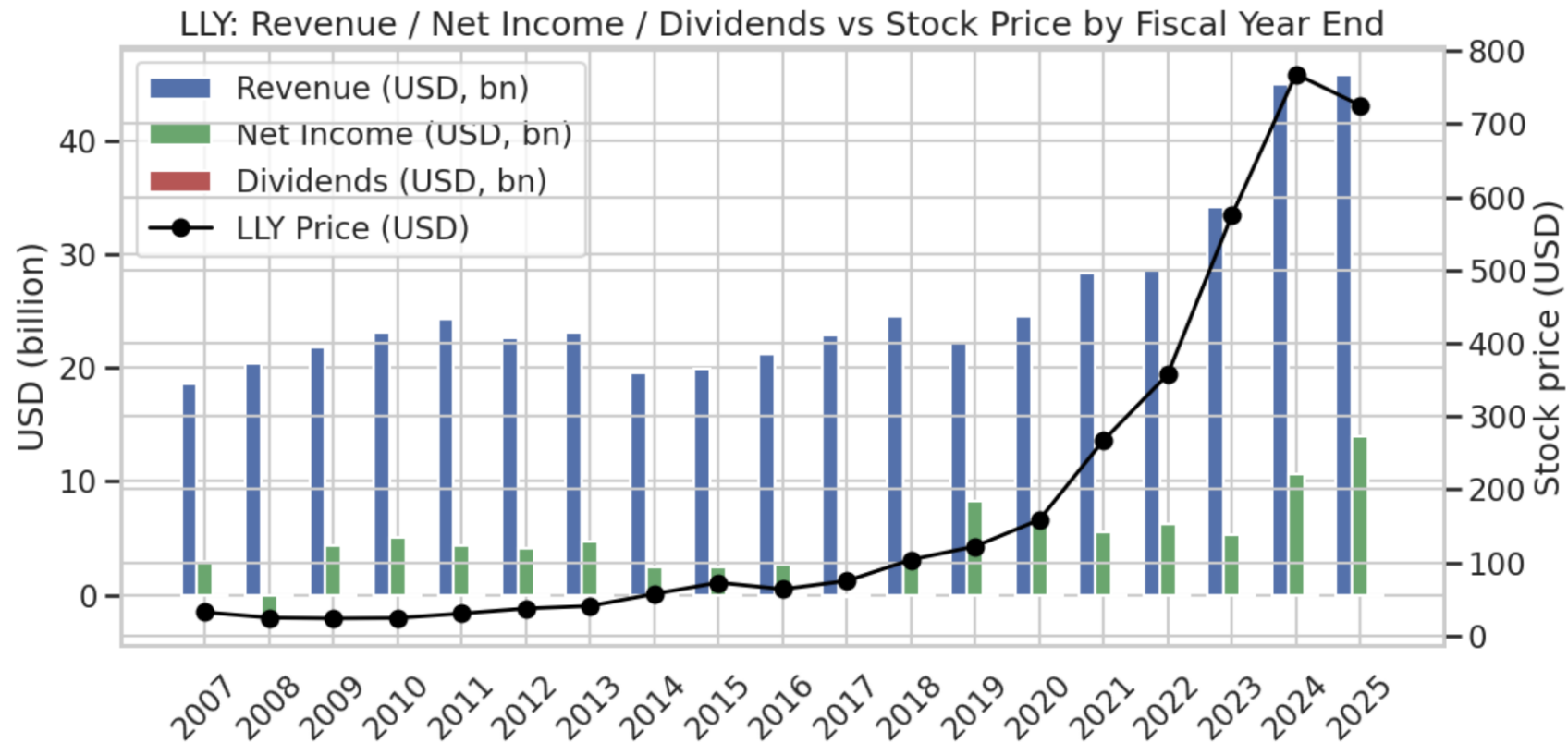
Correlation Heatmap Daily Return Correlations (2023–2025)

- Moderate correlations across Big Pharma
- LLY partially decoupled due to GLP-1 catalysts
- MRK acts defensively



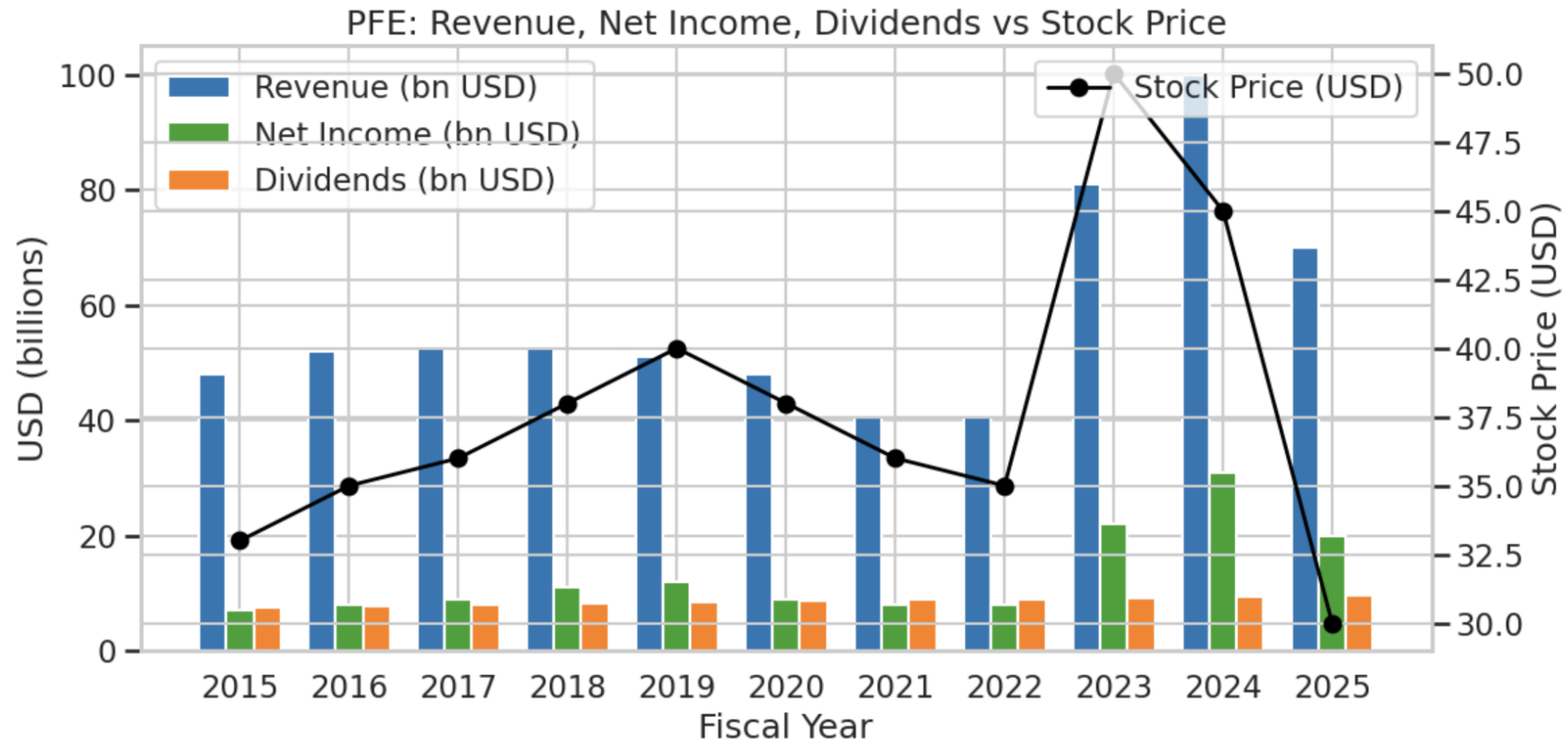
LLY: Revenue / Net Income vs Stock Price

- 2023–2025: strong revenue and profit expansion tied to GLP-1.
- Price rerates in step with fundamentals.



Fundamentals vs Price: Pfizer

- 2021–2022 revenue peak (vaccines) followed by normalization.
- Price de-rates as COVID tailwinds fade.



Key Takeaways

- News tone does not have a strong influence on returns
- Fundamentals are the key drivers of price performance
- New product releases contribute to volatility spikes

Positioning:

Overweight LLY for growth (accept event-driven risk)

Hold MRK as a defensive asset

AMGN for balanced exposure

Be selective with PFE